CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements

Field	Your entry
Project title	North Star Navigation
Team number and section	Section 005, Team 04
Team members (name · net ID)	Joshua Kokes - dal924398 Vincent Jonathan - dal111624 Teddy Byrne - mmb180000 Felipe Gonzalez - dal567737 Aaron Khauv - avk230008 Isaiah Chen - imc210001 Joshua Ayeh-Kwapong - jxa220049 Christian Ermias - cxe210011
Repository URL	https://github.com/sirfindcent/cs3354-team-four
Document version	0.1
Date	October 18, 2026

0 Revision History and Contribution Record

Both tables below are graded. The contribution record carries the individual 20% of this milestone and must be consistent with your repository history.

Version	Date	Author	Summary of change
0.1	10/18/2026		Initial baseline for P1

0.1 Individual contribution record

Team member	What this person produced for P1 (distinguishable contribution, no duplicate row)	Evidence in the repository (a Commit range, an issue number, a pull-request review, a document section you wrote)
Joshua Kokes	Project Name Idea, 1.1 Introduction Statement, 1.3 Definitions. 4.3 Risks	In document
Vincent Jonathan	Product Perspective and Context; Assumption or Dependency; Feasibility Gate; Functional Requirement; Non-Functional Requirement; Constraint; Initial Use-Case Diagram; Requirement to Use Case Traceability;	In document: sections 2.1, 2.5, 2.6, 3.1, 3.2, 3.3, 3.4, 3.5
Teddy Byrne	Project Scope, Constraints, document proofreading	In document: sections 1.2, 2.4
Felipe Gonzalez	Initiated and shared project template, contributed ideas to the project, and participated in brainstorming sessions to develop the project concept.	
Aaron Khauv	Initial idea for an app or website that expands current UTD Map systems. Main voice in early planning of Scope and Requirements.	In document: Sections 1.X, 3.1
Isaiah Chen	Introduced and initiated group communications. Aided in initial ideas, notable stakeholders	In document
Joshua Ayeh-Kwapong	Open Questions, Section 2, constraints, and help with ideas, project scope, and planning.	In document: section 2.3
Christian Ermias	Open Questions, Plan to P2, and Repository and tooling	In document: sections 4.2, 4.4, 5

1 Introduction

1.1 Purpose of this document and of the system

This document and its contents hereafter are the project proposal and specifications for North Star Navigation, a platform for navigating the UTD campus. This document is intended to be submitted to Professor Seker and his TAs for review of the feasibility of the project and the quality of the proposal.

North Star Navigation is primarily targeted towards any person on the UT Dallas campus. The current UT Dallas campus map is poor in its navigational aid as it gives few options for routes, does not offer navigation to specific rooms, and is often time inaccurate. This can lead to students getting lost or arriving in areas later than expected. North Star Navigation alleviates those issues by providing multiple routes to a user that they can choose based on their preferred arrival time and scenery with the inclusion of the interior navigation of every building.

1.2 Scope

In scope — we will build this	Out of scope — we will not; see box below table for why
Website that provides a map of UTD Campus	User accounts
Locations that will be included: UTD Buildings and their indoor room layout, Parking Lots, Bus Stops, Northside, and the Silver Line	Live parking space feed
Active Routing System between Location & Destination	Live weather report
Filtering of Buildings in/out of Routes	Time-dependent event tracking
Building information provided: name, picture of building	DART system information

Out Of Scope:

User accounts: Having user accounts would add privacy concerns and require negotiating with UTD’s IT department. We can't justify that amount of time relative to how much functionality an account system would provide.

Live parking space feed: This is a large enough task to be its own project. We can’t justify that amount of time and effort for what would be a tertiary feature.

Live weather report: Weather apps are plentiful and often come preinstalled on mobile devices. Developing our own would be redundant.

DART system information: DART has a dedicated app for tickets, route planning, etc. Developing our own would be redundant.

Time-dependent event tracking: A log of events taking place on campus requires active upkeep. Not all events that can impede traffic are announced ahead of time, and trying to accommodate around them as they pop up adds a real-time functionality that becomes unwieldy.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
UTD	The University of Texas at Dallas
ECSW	School of Engineering and Computer Science - West

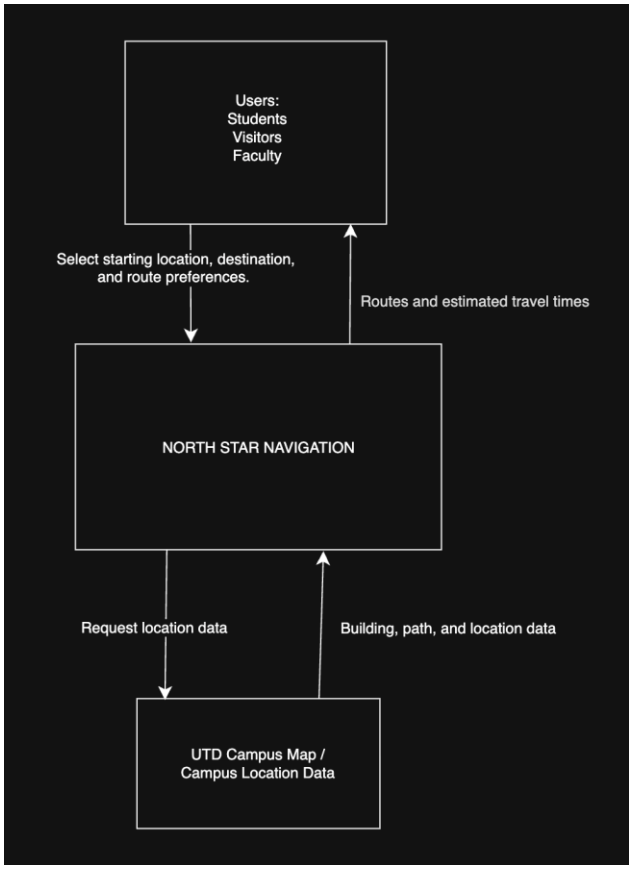
Term or abbreviation	Meaning as used in this project
ECSS	School of Engineering and Computer Science - South
ECSN	School of Engineering and Computer Science - North
ATEC	Edith O’ Donnell Arts & Technology Building
CC, CCH	Canyon Creek Heights
CCHN	Canyon Creek Heights North
CCHS	Canyon Creek Heights South
UV	University Village
Northside	Northside Apartments
Waterview	Waterview Parkway
PS1, PS2, PS3, etc.	Parking Structure 1, Parking Structure 2, Parking Structure 3, etc.
NOTE:	More abbreviations should be added later to account for all of the commonly shorthanded locations on campus.

1.4 References

[R1] UTD Campus Map

2 Overall Description

2.1 Product perspective and context



The system boundary includes North Star Navigation because it is responsible for generating campus routes, estimating travel times, and applying route preferences. Students, visitors, and faculty are outside the system because they use the application by entering a starting location, destination, and route preferences, then receiving routes and estimated travel times. The UTD Campus Map or other campus location data is also outside the system because North Star Navigation depends on external information about buildings, paths, and locations rather than creating all that data itself.

This boundary keeps the project focused on campus navigation and routing instead of taking responsibility for external campus data sources.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Students	The main users/clientele	Estimated travel times	This clientele is the group the system will be developed for.
Visitors	Secondary users	Estimated travel times	Alongside existing students, campus visitors will need to know where they are going.

2.3 User characteristics

Students: The users know how to use navigation apps, basic inputs, understand map symbols/ key. Users cannot be expected to know building nicknames, internal layouts, nor room numbering.
Visitors: Visitors can include highschoolers, transfer student touring the school, and anyone else having a temporary stay at the University. Visitors should know basic map symbols as the minimum. Since they are visiting, they are not expected to know the main building locations, parking, facilities (Recreation Center and Student Union), building nicknames, nor internal layouts.

2.4 Constraints

No user accounts: UTD IT department, adds lead time

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Dependency: The project will have access to reliable location data for UTD buildings, paths, entrances, parking lots, and other supported locations.	If the location data is missing or inaccurate, the routing system may generate incorrect routes, or we may have to reduce the numbers of locations supported.
A2	Assumption: Travel-time estimates can be calculated using route distance and walking speed.	If actual walking times vary too much from the estimate, the travel-time feature may not be accurate enough and would need additional calibration or data.
A3	Assumption: Building and paths used by generated routes will normally be accessible when students use the system.	If a building entrance or path is closed, a generated route could become unusable, and the system would need an alternate route.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible with some risk	The core system is a web-based campus map with route generation and travel time estimates. The current working UTD campus map already **shows** that campus routing is **technically possible**, and our first prototype is limited to ECSW and one nearby parking lot before expanding.
Operational	Feasible	The system will run as a **website** and will not require user accounts. Users only need to select a starting location, destination, and optional route preferences, which keeps the system **simple** to operate.
Schedule	Feasible if scope stays limited	The project is planned in **small iterations**: first ECSW and one parking lot, then the ECS area, then one longer cross-campus route. More complex features such as full indoor navigation can be **postponed** if needed.
Economic / resource	Feasible	The project does not require physical equipment or paid infrastructure for the initial version. The **main resources** are team development time, campus location data, and standard web-development tools.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-02	The system shall allow a user to select a starting location from the supported campus locations.	MUST	Students / Visitors	A tester can select a valid destination and see it accepted by the system.
FR-03	The system shall generate at least one route between the selected starting location and destination.	MUST	Students / Visitors	A tester can choose two supported locations and receive a route connecting them.
FR-04	The system shall display an estimated travel time for each generated route.	MUST	Students / Visitors	A tester can generate a route and see an estimated travel time displayed.
FR-05	The system shall display an estimated travel time for each generated route.	MUST	Students / Visitors	A tester can generate a route and see its estimated travel time
FR-06	The system shall allow a user to apply route preferences when generating a route.	SHOULD	Project Scope	A tester can select at least one route preference and generate a route using it.
FR-07	The system shall allow a user to avoid selected buildings or paths when an alternative route exists.	SHOULD	Project Scope	A tester can exclude a supported building or path and receive an alternate route.
FR-08	The system shall display basic information for supported campus locations.	SHOULD	Project Scope	A tester can select a supported location and view its available information.

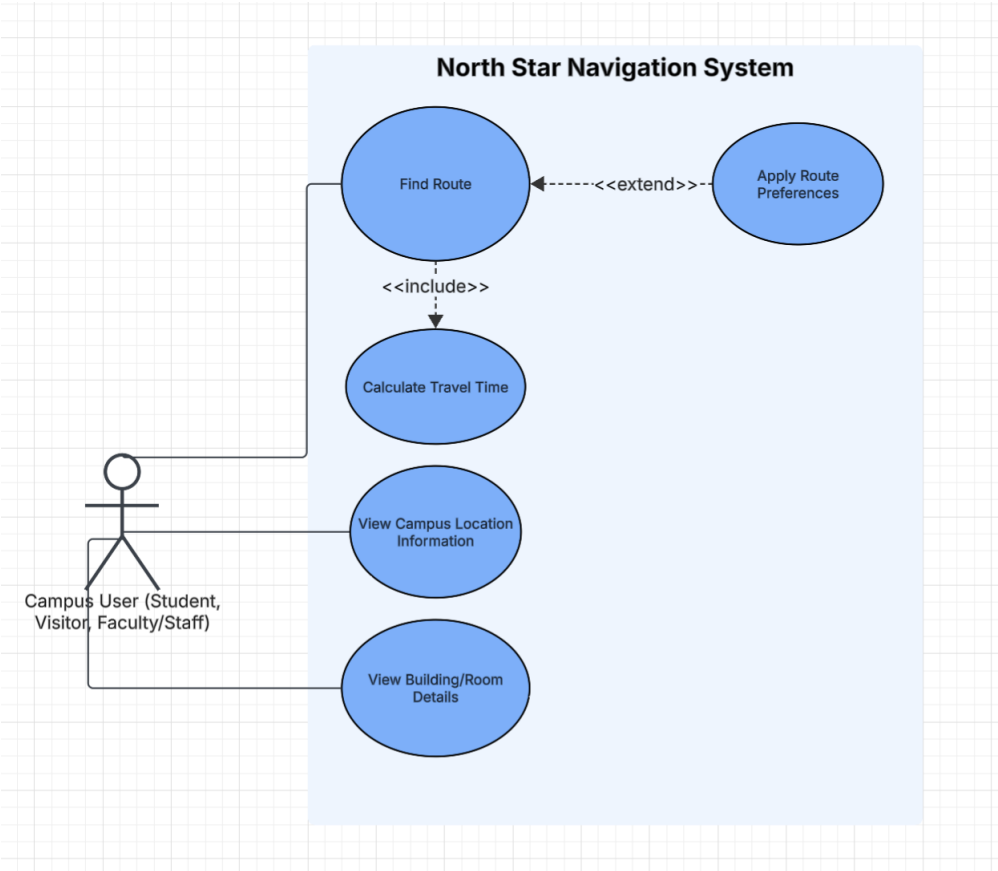
3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	The system shall display a generated route within **3 seconds** after the user selects a valid starting location and destination under normal use.	MUST	Students / Visitors	Measure the response time for multiple route requests and confirm they complete within 3 seconds.
QR-02	The system shall display travel-time estimates within **20% of the measured walking time** for tested campus routes.	SHOULD	Students / Visitors	Walk selected test routes, record the actual times, and compare them with the system estimates.
QR-03	The system shall successfully generate a route for at least **95% of valid route requests** between supported locations.	MUST	Students / Visitors	Test a set of valid origin-destination pairs and calculate the percentage that return a route.
QR-04	The system shall allow a first-time user to generate a route in **3 steps or fewer** after opening the map.	SHOULD	Students / Visitors	Give the system to a first-time user and count the number of actions needed to generate a route.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall operate **without requiring user accounts**.	Project scope	User accounts are explicitly outside the project scope and would add privacy and authentication work.
CON-02	The system shall use **existing UTD campus/location data or other approved external location data** rather than requiring the team to manually survey the entire campus.	Project scope	The project depends on existing location information to stay feasible within the project schedule.

3.4 Initial use cases



UC-01 Find Route

Field	Content
Identifier and name	**UC-01 Find Route**
Primary actor	Campus User
Supporting actors	UTD Campus Map / Campus Location Data

Field	Content
Preconditions — what must be true before it starts	The system has access to supported campus location data.
Trigger — what starts it	The user selects a starting location and destination and requests a route.
Main success scenario — numbered steps	1. The user selects a starting location. 2. The user selects a destination. 3. The system retrieves the needed campus location data. 4. The system generates a valid route. 5. The system calculates the estimated travel time. 6. The system displays the route and estimated travel time.
Alternative and exception flows	If no valid route exists, the system informs the user. If required location data is unavailable, the system informs the user that routing cannot be completed.
Postconditions — what is true when it succeeds	The user is shown a valid route and its estimated travel time.
Requirements realized	FR-01, FR-02, FR-03, FR-04, FR-05

UC-02 Apply Route Preferences

Field	Content
Identifier and name	UC-02 Apply Route Preferences.
Primary actor	Campus User
Supporting actors	None
Preconditions — what must be true before it starts	The user has selected a valid starting location and destination.
Trigger — what starts it	The user selects one or more available route preferences.
Main success scenario — numbered steps	1. The user selects a route preference. 2. The system applies the preference to the route search. 3. The system generates a route that follows the preference when possible. 4. The system calculates the travel time. 5. The system displays the route and estimated travel time.
Alternative and exception flows	If no route satisfies the selected preference, the system informs the user. The user may remove or change the preference and try again.
Postconditions — what is true when it succeeds	The user receives a route that follows the selected preference when a valid route exists.
Requirements realized	FR-06, FR-07

3.5 Requirements to use case traceability (Vincent)

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	**UC-01 Find Route**	-
FR-02	**UC-01 Find Route**	-

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-03	**UC-01 Find Route**	-
FR-04	**UC-01 Find Route**	-
FR-05	**UC-01 Find Route**	-
FR-06	UC-02 Apply Route Preferences	-
FR-07	UC-02 Apply Route Preferences	-
FR-08	**UC-03 View Campus Location Information** and **UC-04 View Building/Room Details**	-

The matrix showed that all functional requirements are connected to at least one use case, so there are no unimplemented functional requirements.

It also confirmed that each use case is supported by at least one functional requirement. No changes were required, but the matrix helped verify that the requirements and use cases are consistent with each other.

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Vincent Jonathan	Context model, assumptions/dependencies, feasibility, requirements review	Same team decision process
Teddy Byrne	Scheduling meetings, software architecture design	Meeting with disagreeing member(s) to negotiate; pull in other team members as necessary; majority rule as last resort
Joshua Kokes	Software architecture design, API implementation	Step through and weigh options between team members

4.2 Plan to P2

Date	What will be finished	Owner
Oct. 22	Review P1 feedback and revise incomplete or unclear requirements.	Christian Ermias, Joshua Ayeh-Kwapong
Oct. 26	Finalize functional requirements, quality requirements, and use cases.	Vincent Jonathan
Oct. 30	Develop the software architecture and identify the major system components.	Teddy Byrne, Joshua Kokes

Date	What will be finished	Owner
Nov. 4	Create class diagrams and sequence diagrams illustrating the system's design.	Felipe Gonzalez, Aaron Khauv, Isaiah Chen
Nov. 8	Review, finalize, and submit the completed P2 document.	Teddy Byrne

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	Web server going down	Medium	Very High	Ensure systems operations has routine maintenance by team
R2	Incorrect arrival time estimations	High	Low	The team will collect large enough dataset for estimation
R3	Routing to incorrect destination	Low	Very High	The team must ensure proper testing is done on the routing system before deployment
R4	Route system not following preferences set	Medium	High	Propper testing by the team is needed to prevent this case
R5	Navigation route is blocked due to external circumstances (ex. Construction)	High	High	Live campus updates is out of scope for the project, may implement later

4.4 Repository and tooling

Our team will use GitHub to manage and organize the North Star Navigation project.

Repository: https://github.com/sirfindcent/cs3354-team-four

Branch conventions: The main branch will contain approved project work. Team members will create separate branches for their individual contributions before merging changes.

Commit conventions: Commit messages will briefly describe the changes made to the project.

Issue tracking: GitHub Issues will be used to assign tasks, track progress, and document problems that need to be resolved.

Repository access: All contributing team members will be given access to the repository, and the instructor will be able to view the project.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
ChatGPT	Helped brainstorm and review functional requirements, quality requirements, constraints, use cases, context diagrams, and wording for project sections.	Vincent Jonathan	The output was compared against the course slides/template, checked against the project scope, and manually revised before being added to the document.

Tool	What it was used for	Who used it	How the output was checked
ChatGPT	Assisted with drafting project milestones, repository conventions, and open questions.	Christian Ermias	Reviewed against the assignment template and project requirements, then manually revised.

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	How will we determine whether our generated routes and estimated walking times are sufficiently accurate?	Development team	Oct. 25	We will initially evaluate routes by checking whether they reach the correct destination and comparing estimated travel times with actual walking times.
Q2	How will the navigation system handle pathways that become newly blocked or inaccessible?	Development team	Oct. 25	We will initially rely on existing campus pathway data and will not support real-time blockage updates.
Q3	Which route preferences can we feasibly support within the project timeline?	Development team	Oct. 25	We will initially prioritize shortest routes and allow users to avoid selected buildings or pathways.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Joshua Kokes	Joshua Kokes	10/18/2026
Vincent Jonathan	Vincent Jonathan	10/18/2026
Teddy Byrne	Teddy Byrne	10/18/2026
Felipe Gonzalez	Felipe Gonzalez	10/18/2026
Aaron Khauv	Aaron Khauv	10/18/2026
Isaiah Chen	Isaiah Chen	10/18/2026
Joshua Ayeh-Kwapong	Joshua Ayeh-Kwapong	10/18/2026
Christian Ermias	Christian Ermias	10/18/2026